

ClassMic Integrated Project Plan

D1 to D7 Course prototype and two weekly sprints

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D1 Development approach

Use a hybrid approach. Complete the D planning outputs in sequence, then build and review the prototype in two weekly E sprints. Close the course project in F using the actual results. The revised A2, B2 and C1 define the proposed scope. This plan is ready for review, but it does not claim stakeholder validation or charter approval.

Predictive planning	Adaptive construction
Agree on the classroom problem, five-story scope, responsibilities and acceptance boundary before development.	Use Sprint 1 to test the session and instructor workflow, then revise priorities from feedback.
Set the course milestones, effort forecast, risk responses and test approach. Confirm permission and data rules.	Use Sprint 2 for the audio path, recovery, control tests and evaluation. Correct defects before optional polish.
Record approvals and changes. Keep the D baseline stable enough to track progress.	Update the Notion PRD and backlog after each review while keeping the charter objectives traceable.

Why the approach fits

The assessed outputs and deadlines are fixed, so the team needs a plan before building. The audio path, browser behavior and instructor workflow are uncertain, so short demonstrations and feedback should guide implementation. Two students can coordinate through a small backlog without creating separate management roles for every Scrum responsibility.

Controls and assumptions

No live classroom rollout is included. Controlled tests need willing participants, available equipment and an agreed privacy approach. The course instructor reviews material scope or schedule changes. Any university room test or new spending requires the relevant permission before it takes place.

The current brief names D7 in its overview but supplies detailed prompts only for D1–D6. The D7 review in this document checks consistency across those outputs. D1 remains unvalidated until the team and instructor record their review.

Decision to record

The selected baseline is five stories, 100 student hours and the two listed sprint dates. Formal course review and actual resource availability remain open. The older 14-week, three-room pilot is a possible later project and is outside this plan.

D2 Scope statement

ClassMic will test whether a student phone can carry a question through a speaker after instructor approval. The project delivers a PRD and backlog, a browser prototype covering five user stories, controlled test evidence, volunteer feedback and a recommendation. It includes temporary session access, microphone permission, speaking requests, one approved push-to-talk microphone, instructor stop controls and a recovery path.

The team will not deliver a production service, run a six-week live classroom pilot, save audio or transcripts, identify speakers by voice, grade participation or support simultaneous microphones. It assumes access to existing devices and a speaker, approximately 100 student hours and willing reviewers. Completion requires all Must acceptance checks, the charter's proposed evaluation measures, an honest account of failures and no open critical control defect. Course deadlines and two weekly construction sprints constrain delivery.

Requirements summary

ID	Requirement	Charter link
R1	Temporary browser session and QR or code join	O1; O2; O4
R2	Speaking queue with approval, mute, removal and end controls	O1; O4
R3	One approved phone transmits to a laptop and connected speaker	O1; O2; O3; O4
R4	Clear permission, connection and alternative participation paths	O1; O4
R5	Test record and anonymous feedback without stored speech	O1; O2; O3; O4

PRD structure and ownership

Nurtore maintains the PRD. Temiko updates implementation status and test evidence. The supplied Notion import table uses the required fields: Epic, Feature/Requirement, User Story, Persona, Priority, Acceptance Criteria, Charter Objective and Status. The separate PRD text contains the context, personas, scope and open questions.

Until the team imports the files and provides the page link, the Notion requirement remains open. After import, use that page as the maintained requirements record and retain exports at review milestones.

Evidence boundary

R1–R3 follow the in-class ClassMic concept and prior charter. R4 and R5 carry forward access and evaluation needs from the prior drafts. Each remains a proposed requirement until students and an instructor review it. No interview, survey or usability result has been invented.

D2 Acceptance and priorities

Story	Acceptance boundary	Iteration
US1 / R1	Join the intended session. Reject expired access. Handle denied permission. Meet the setup target.	Sprint 1
US2 / R2	Queue once, approve one speaker, and stop audio through mute, removal, timeout and end. Pass misuse tests.	Both
US3 / R3	Transmit only with permission, approval and push to talk. Meet the speaking-start target. Measure audio delay separately.	Sprint 2
US4 / R4	Explain denied permission and connection loss. Preserve an alternative speaking path and clear controls.	Both
US5 / R5	Record all acceptance results and actual feedback counts. Save no speech. Report defects and limits.	Sprint 2

Priority decisions

Must: R1–R5 form the minimum complete prototype and evaluation. A failed core control, audio path or evidence requirement cannot be hidden by marking the prototype complete.

Should: simple connection checks and a short operator guide, where they help reviewers use the prototype. These refine the existing stories and must not delay acceptance tests.

Could: visual polish after the core workflow passes. Any new capability needs a requirement and evidence before it enters a sprint.

Will not have: recording, transcript storage, voice identification, attendance, grading, persistent profiles, captions, polling, remote participation or campus deployment in this course release.

Open questions for validation

Which classroom situations cause audibility problems, and how often? Will instructors operate a queue during discussion? Which devices and network can the team test? What delay is acceptable with the available speaker? Which volunteer reviewers can participate? What minimum metadata, access and deletion rules will the team use?

Traceability and change control

Every backlog entry carries R1–R5 and O1–O4 references. Each test result names its requirement, method, device and outcome. Nurtore records requested changes and their effect on the scope, effort and course deadline. The instructor reviews material changes before the team treats them as approved.

D3 Work breakdown structure

The WBS contains 18 elements, including six summary elements. Each leaf has an owner and an identifiable output. Requirements R1–R5 map to the five user stories in the PRD.

WBS	Element	Output	Owner
1.0	Project authorization and control	Summary	Nurtore
1.1	Reviewed A2, B2 and charter	Decision and confirmation record	Nurtore
1.2	Status and change records	Current actions, risks and decisions	Nurtore
2.0	Requirements and plan	Summary	Nurtore
2.1	Stakeholder input and PRD	Five validated or explicitly open stories	Nurtore
2.2	Integrated D plan	Scope, schedule, risk, quality, cost and team plan	Nurtore
3.0	Sprint 1 increment	Summary	Temiko
3.1	Session access and permissions	R1 working flow	Temiko
3.2	Instructor queue and controls	R2 control flow and review	Temiko
4.0	Sprint 2 increment	Summary	Temiko
4.1	Audio path and recovery	R3 and R4 integrated prototype	Temiko
4.2	Acceptance and evaluation	R1–R5 test record and feedback	Nurtore
5.0	Delivery evidence	Summary	Nurtore
5.1	E demonstration	Working demo and defect record	Temiko
5.2	Updated requirements record	PRD with links to tests and decisions	Nurtore
6.0	Closing	Summary	Nurtore
6.1	Outcome and lessons review	Actual results against O1–O4	Nurtore
6.2	Final presentation and handover	F presentation and retained project files	Nurtore

Required human input: the team must supply its own hand-drawn or Post-it WBS and dependency network. This typed WBS is a planning proposal and does not replace those images.

D3 Schedule and release plan

All dates below are proposed. Durations are calendar-day windows, not full-time labor. The plan completes D early so that two full weekly sprints fit before E is due. D needs review by 19 September. The E upload area opens on 26 September; its opening date does not prohibit earlier work.

ID and activity	Dates	Days	Depends on
S1 A2, B2 and charter review	10 Sep–12 Sep	3	None
S2 Requirements and stakeholder input	13 Sep–15 Sep	3	S1
S3 Technical approach and device access	13 Sep–15 Sep	3	S1
S4 Integrated D plan and review	16 Sep–19 Sep	4	S2;S3
S5 Sprint 1 session and control workflow	20 Sep–26 Sep	7	S4
S6 Sprint 2 audio integration and evaluation	27 Sep–03 Oct	7	S5
S7 Closeout and final presentation	04 Oct–09 Oct	6	S6

Gantt view

Activity	10–12 Sep	13–19 Sep	20–26 Sep	27 Sep–3 Oct	4–9 Oct
A2, B2 and C1 review	Work				
Requirements and D review		Work			
Sprint 1			Work		
Sprint 2 and E				Work	
Closing and F					Work

Critical dependencies and resource conflicts

S1 precedes parallel S2 and S3. Both must finish before S4, then S5, S6 and S7 follow in order. These are controlling paths with no planned buffer before the internal milestone dates. E is due at 00:00 on 4 October, so finish and check the upload on 3 October. F follows the same pattern on 9 October.

Audio feasibility and Temiko's availability have low-confidence estimates. Nurtore's testing and evidence work overlap Sprint 2. Reserve review time in each sprint. If D approval slips past 19 September, ask the instructor to resolve the two-week schedule conflict. Do not silently shorten either sprint or assume an extension.

D4 Risk register

The register has six threats followed by two opportunities. Probability and impact are qualitative estimates awaiting team review. High impact means a core objective, deadline or control could fail. Medium impact means rework or weaker evidence. Low impact means a local inconvenience.

ID	Event	Prob	Impact	Response	Owner
T1	Speaker feedback or delay makes questions hard to understand.	High	High	Test the audio path early at low volume. Adjust the setup and retain a fallback.	Temiko
T2	Browser or network differences prevent some devices from joining or transmitting.	Medium	High	Test available devices and permissions early. Publish the tested combination.	Temiko
T3	A control or configuration defect permits unwanted capture or transmission.	Medium	High	Keep one approved speaker. Test denial, expiry, mute and end. Save no speech.	Temiko
T4	Competing coursework or late D approval leaves too little time for two sprints.	High	High	Confirm capacity and the early review date. Reserve testing time and drop optional work.	Nurtore
T5	Unavailable or excluded reviewers leave too little useful feedback.	Medium	Medium	Invite willing reviewers early. Provide an alternative speaking path and report sample limits.	Nurtore
T6	Unclear audio expectations or extra features expand the work after planning.	Medium	High	Agree on acceptance before building. Assess changes against the five-story boundary.	Nurtore
OP1	Existing equipment or adviser support becomes available for a better sound test.	Medium	Medium	Seek access and a short review without making it a delivery dependency.	Nurtore
OP2	Standard browser capabilities work well enough to reduce integration effort.	Medium	Medium	Prototype the transport early and reuse supported components after review.	Temiko

Risk ownership

Owners review triggers during the team check-in and at each sprint review. Record a risk as an issue when it occurs. Nurtore escalates a threatened course deadline or scope change to the instructor. Temiko stops an affected test immediately if microphone control fails.

D4 Triggers and contingency actions

ID	Observable trigger	Contingency or opportunity action
T1	Sustained feedback or listeners cannot follow the test speech.	Stop audio. Retest at safe settings or use another speaker setup. Report an unmet audio requirement if unresolved.
T2	A required test device cannot join or transmit after troubleshooting.	Use the documented supported device for the demo and record the compatibility failure. Do not claim wider support.
T3	Any unapproved transmission or saved speech appears.	Stop the test and disable the affected path. Correct the defect and rerun misuse tests before resuming.
T4	D lacks approval by 19 September or a core sprint item exceeds available effort.	Escalate dates or scope for instructor review. Remove optional polish. Preserve test and evidence time.
T5	Fewer than five willing reviewers are available or a reviewer lacks an equivalent participation route.	Arrange another controlled review or report insufficient evidence. Never invent responses or force participation.
T6	A request adds a new feature or changes the agreed acceptance threshold.	Log impact and obtain a decision before adding work. Keep the original result visible if a target changes.
OP1	An adviser or existing room setup is available without new spending.	Schedule an additional comparison test with permission. Record the configuration and findings.
OP2	Early transport checks pass with less effort than estimated.	Use the released time for device and accessibility testing. Do not expand the release boundary automatically.

Response records

For each triggered event, record the date, evidence, owner, action and effect on the next milestone. Close it only when the response has been checked. Opportunities count as benefits only when they occur and the team records what changed.

D4 Quality and acceptance plan

Quality assurance prevents avoidable defects: review requirements before coding, check changes against the PRD, review microphone data flow and keep an agreed definition of done. Inspection and testing check the resulting prototype: run functional, permission, misuse, audio and user-review scenarios and record their outcomes.

Test area	Acceptance evidence	Owner
Scope and controls	R1–R5 checklist. All Must checks pass. Tests cover two simultaneous requests, mute, removal, expiry, disconnect and end.	Both
Setup and speaking start	10 setup trials, at least 9 within 2 minutes. 30 ready-speaker trials, at least 27 audible within 3 seconds of approval.	Nurtore
Audio quality and delay	Measure 30 one-way delay trials. Report median and 95th percentile; the latter must be at most 150 ms. Test intelligibility and sustained feedback. Record any failure.	Temiko
Privacy and data	Configuration and storage review shows no saved speech. Misuse tests show no unauthorized transmission. Save minimal test metadata only.	Temiko
Access and recovery	Permission denial, reconnect and alternative participation checks. Review readable labels and keyboard access.	Both
User feedback	At least 5 willing reviewers compare audibility with the current workaround. At least 75% positive, with counts, response wording and limitations reported.	Nurtore

Measurement protocol

Start setup timing when the instructor opens a new session and end when a test student is ready to request a turn. Use a scripted ready speaker for activation timing. For transport delay, use an agreed synthetic test signal and one calibrated timing method, not unsynchronized phone clocks. Retain timing results only. Note the device, browser and network for every trial.

Ask reviewers whether questions were easier to hear than through the baseline workaround. Use the same test setting and explain the small convenience sample. With five responses, four positive responses are 80% and meet the proposed 75% target. This does not establish a campus-wide effect.

Defects and acceptance

Critical defects include unwanted transmission, saved speech or a failed stop control. Stop affected testing and require a verified fix. Other defects need an owner, priority and retest result. A failed Must check remains open in the report. The team reviews the evidence and the instructor accepts the assessed deliverable. Record actual outcomes during E.

D5 Cost and resource plan

The course prototype uses a provisional estimate of 100 student hours. These are planning allowances, not recorded time. Nurtore tracks actual effort and the forecast after each sprint. No wage rates or supplier quotations were supplied.

Work	Temiko hours	Nurtore hours	Total hours
Initiation and D planning	18	22	40
Sprint 1	18	6	24
Sprint 2	14	10	24
Closing and presentation	4	8	12
Total hours	54	46	100

Resource	Planning basis	Confirmation needed
Phones, laptop and speaker	Reuse available equipment. Incremental cash allowance AED 0.	Availability and compatibility. Any purchase changes the estimate.
Development and hosting	Use already available tools and an approved local or existing environment.	Permitted environment and access. No paid subscription assumed.
Notion PRD	Prepare importable PRD content and the required database fields.	Existing team account or page and access preferences.
Volunteer and adviser time	Participation by agreement. No cash allowance or external commitment.	Willing reviewers and any technical advice.

Cash baseline and uncertainty

Provisional new cash spending is AED 0 if all assumed resources are available. This is a reuse condition, not evidence that a production system costs nothing. Any paid requirement must have a quote, an owner and approval before commitment. The previous AED 60,000 pilot ceiling does not apply to this course prototype.

The effort estimate has low confidence until the early audio test. Temiko's 18-hour Sprint 1 allowance and 14-hour Sprint 2 allowance are the main capacity limits. Reserve Nurtore's evaluation time rather than shifting all testing to the final evening.

Lower resource scenario

If total effort falls to about 70 hours, first remove polish and extra device coverage. Keep the permission, one-speaker, stop-control and no-recording requirements. Seek approval for a reduced evaluation scope if necessary. A click-through mockup can still explain the workflow, but it cannot count as passing live-audio acceptance. Record that change instead of claiming the original scope was delivered.

D6 Team communications and engagement

Role	Proposed responsibility and authority
Nurtore Arynuruly	Project manager and requirements lead. Maintain PRD, schedule, risk and decision records. Coordinate reviews and evaluation.
Temiko Machavariani	Technical lead. Implement and test the prototype. Maintain technical evidence and stop tests for control failures.
Course instructor	Proposed course sponsor and acceptance authority. Review material scope and schedule changes. Confirm name and role.
Student and instructor reviewers	Volunteer feedback on audibility, access and controls. No participation is assumed.

Working agreements

Keep requirements in the agreed Notion PRD after import. Name one owner for each active task and link its evidence. Review changes that affect microphone behavior before merging or demonstrating them. Record meeting decisions and unresolved disagreements. Do not attribute AI-written material or planned feedback to a person as a completed contribution.

Audience and purpose	Proposed cadence	Channel and owner
Team actions, blockers and risks	Ten-minute check-in on workdays; a 30-minute review at each sprint boundary.	Telegram for daily coordination; Notion for decisions. Nurtore keeps the action log.
Sprint planning and review	At the start and end of each weekly sprint.	Working demo and PRD update. Both attend.
Instructor decisions	Before Sprint 1 and when scope or deadlines are threatened.	LMS messaging or the next class review. Nurtore prepares the decision request.
Volunteer feedback	After consent and at the agreed test session.	Use the supplied anonymous review form after a controlled demonstration. Nurtore records results.
Technical or access advice	Before a room test, if an adviser is available.	Agreed review route. Temiko records the recommendation.

Escalation and engagement

Stop affected testing immediately for unwanted transmission or saved speech. Raise an unresolved Must defect, unavailable resource or threatened milestone at the next team check-in and promptly seek the instructor's decision when it changes scope or timing. Ask priority reviewers how and when they prefer to participate. Collect no private contact details in the public project record.

The companion brainstorming supplement explains the selected communication plan. It is AI-assisted planning prepared now. The D6 request for original pre-AI human notes remains open; this supplement does not claim to recreate those notes.

D7 Integrated plan review

This review links the planning outputs and identifies the remaining decisions. It does not mark the project approved or the prototype tested.

Check	Current position	Next action
One project across A–D	All revised documents use ClassMic and the two-sprint course prototype.	Use the course prototype as the working scope; obtain formal course acceptance.
Business case to charter	GO applies to a bounded prototype. A funded live pilot needs another decision.	Record A2 B2 and C1 review.
Scope to backlog	Five Must stories map to O1–O4. Detailed acceptance fields are ready for Notion.	Import PRD and record the page link.
Backlog to WBS	18 WBS elements cover planning, two increments, evidence and closing.	Add genuine WBS and dependency images.
Schedule to capacity	Two seven-day sprints require D review by 19 September. Effort totals 100 hours.	Complete D review by 19 September and check available effort at sprint planning.
Risks to quality	Six threats and two opportunities have triggers, responses and owners.	Track the risk triggers and apply the selected 150 ms delay target.
Resources to cash	AED 0 incremental cash assumes reuse. There are no quotes or paid commitments.	Verify access or revise the estimate before spending.
Team to evidence	Responsibilities and communication proposals are documented.	Supply pre-AI communication notes and charter discussion evidence.

Approval record to complete

Review decision and conditions: _____

Course instructor or agreed approver: _____ Date: _____

Project manager: _____ Date: _____

Requirements used for this revision

The current D Phase Labs attachment controls the D1 one-page limit, D2 two-page limit outside Notion, two weekly sprints, no more than 20 WBS elements, and no more than eight risk events including two opportunities. These replace the older generic planning quantities. The attachment refers to C2 even though the visible charter assignment requests C1; instructor confirmation is recorded as an open item.

[D assignment and current attachment](#)

[E execution assignment](#)

[F closing assignment](#)